

Estatística (bayesiana) aplicada ao monitoramento de doenças infecciosas no Brasil

Leo Bastos



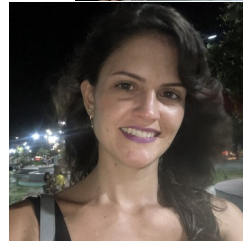
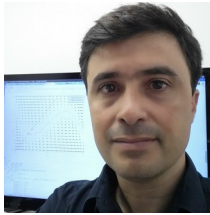
Minha história com o DEST/UFMG

- **Entrei na graduação em estatística em 1998**
 - “Bolsista” FUMP (Fundação Mendes Pimentel)
 - Fui monitor de Probabilidade e Introdução à Estatística
 - PIBIC-CNPq (Rosangela Loschi) trabalhando com MPP
 - Fui aceito no mestrado (2002), optei ir para UFRJ
- **Participei de algumas bancas de mestrado e doutorado**
- **Medalha de Honra UFMG 2024 (Obrigado pela indicação!)**



Collaboration and funding related projects

PROCC/Fiocruz and EMap/FGV



INI/Fiocruz (COVID-19 Cohorts)

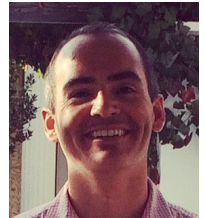
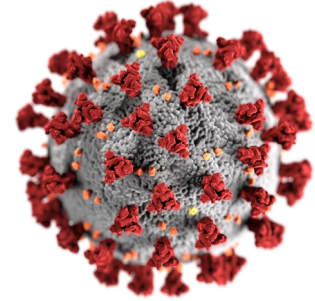
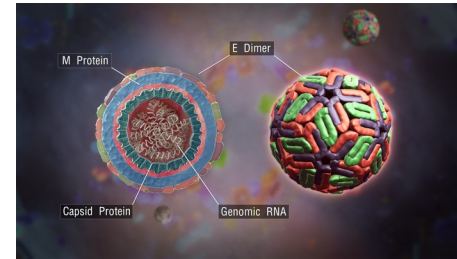


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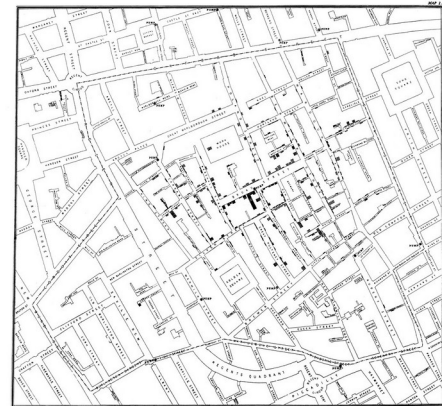
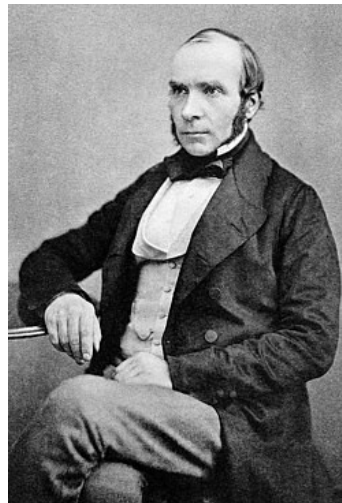
- **Bayes & Epidemiology**
- **COVID-19 cohort studies**
 - Household transmission, reinfection, etc
- **Administrative data (Dengue and SARI)**
 - Contrafactual analysis (Vaccination impact)
 - Early warning systems (Nowcast and Forecast)
 - Defining epidemic thresholds



When Bayes meets John Snow



- **Thomas Bayes (1701-1761)**
- $p(\theta \mid x) \propto p(\theta)L(\theta; x)$
- **Father of Bayesianism**

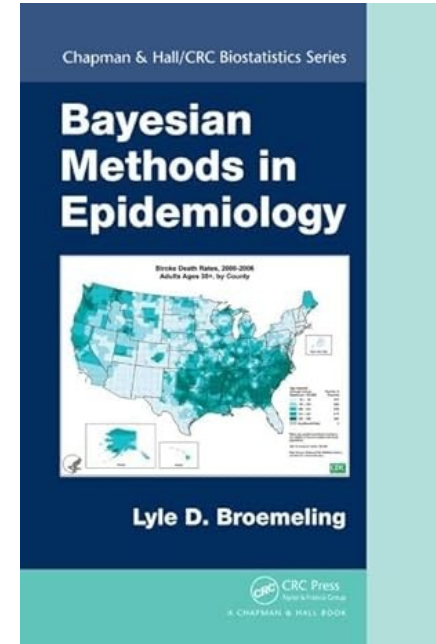
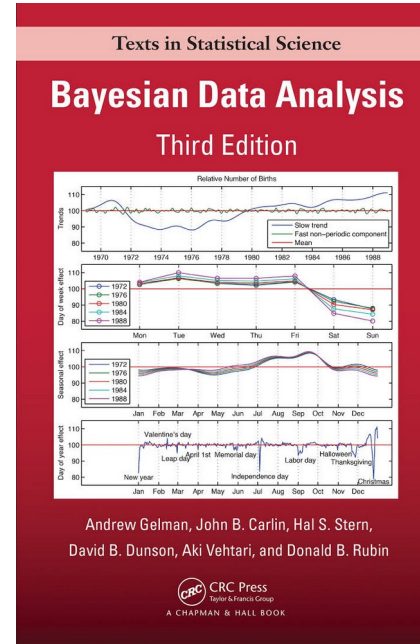
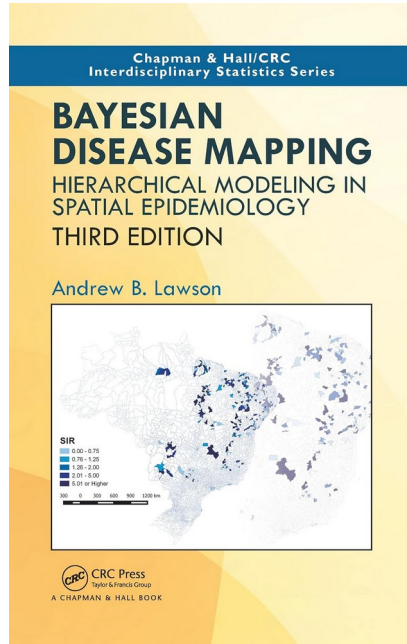
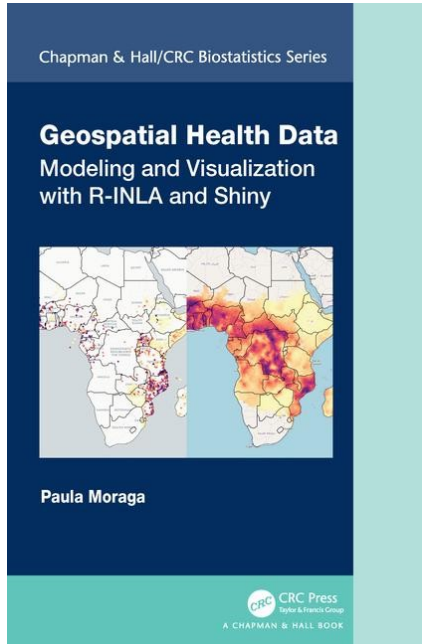


- **John Snow (1813-1856)**
- **Studied cholera in London**
- **Father of epidemiology**

When Bayes meets John Snow



When Bayes meets John Snow



Generalised linear mixed models

Likelihood $Y_i \mid \mu_i, \phi \sim F(\mu_i, \phi), \quad i = 1, 2, 3, \dots, N$

$$g(\mu_i) = \alpha + x_i^T \beta + \gamma_{R[i]}.$$

Prior $\alpha \sim N(0, V_\alpha) \quad \beta \sim N(0, V_\beta) \quad \{\gamma_r\} \sim N(0, Q_\gamma)$

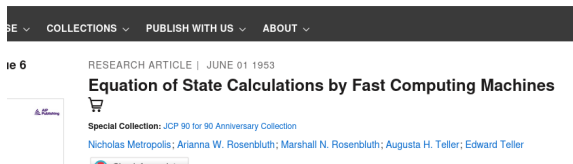
Posterior $p(\alpha, \beta, \{\gamma_r\}, \phi \mid y_1, y_2, \dots, y_N)$



The inference/learning process with respect to the parameters is based on the posterior.

Timeline of Bayesian computation

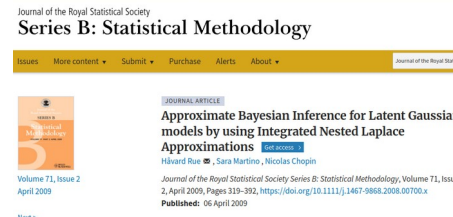
The Journal of Chemical Physics



Metropolis et al. (1953)



Gelfand & Smith (1991)



Rue, Martino & Chopin (2009)

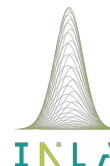
Hastings (1970)



WinBUGS (1997)



JAGS (2007)



STAN (2012)



Modelling household cohorts

Open access

Cohort profile

BMJ Open Cohort-profile: Household transmission of SARS-CoV-2 in a low-resource community in Rio de Janeiro, Brazil

Patricia Brasil ¹, Luana Damasceno,¹ Trevon Fuller,^{1,2} Leonardo S Bastos,³ Oswaldo G Cruz,³ Fernando Medeiros,¹ Guilherme Amaral Calvet ⁴, Paola Resende,⁵ Jimmy Whitworth,⁶ Chris Smith ^{7,8}, Marilda M Siqueira,⁹ Marilia Carvalho³

Brasil et al (2022)

CSP CADERNOS DE SAÚDE PÚBLICA
REPORTS IN PUBLIC HEALTH 

Cohort profile: follow-up of a household cohort throughout five epidemic waves of SARS-CoV-2 in Rio de Janeiro, Brazil

da Silva et al. (2024)

Participants entered in the cohort and their families were followed for some time

Household transmission of SARS-CoV-2 in Manguinhos

Families: 845
Participants: 1760
RT-PCR: 7068; Serologies: 5432

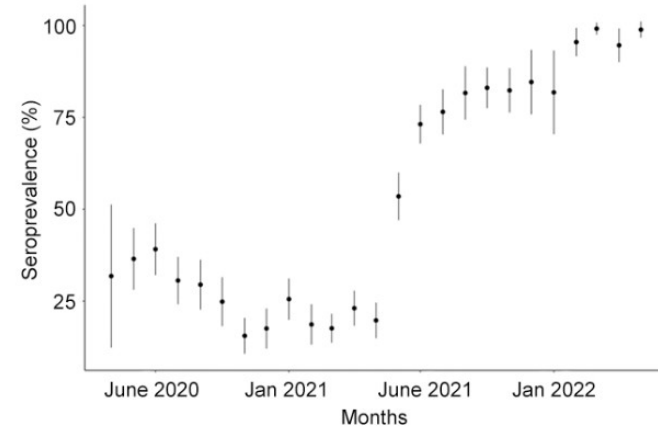
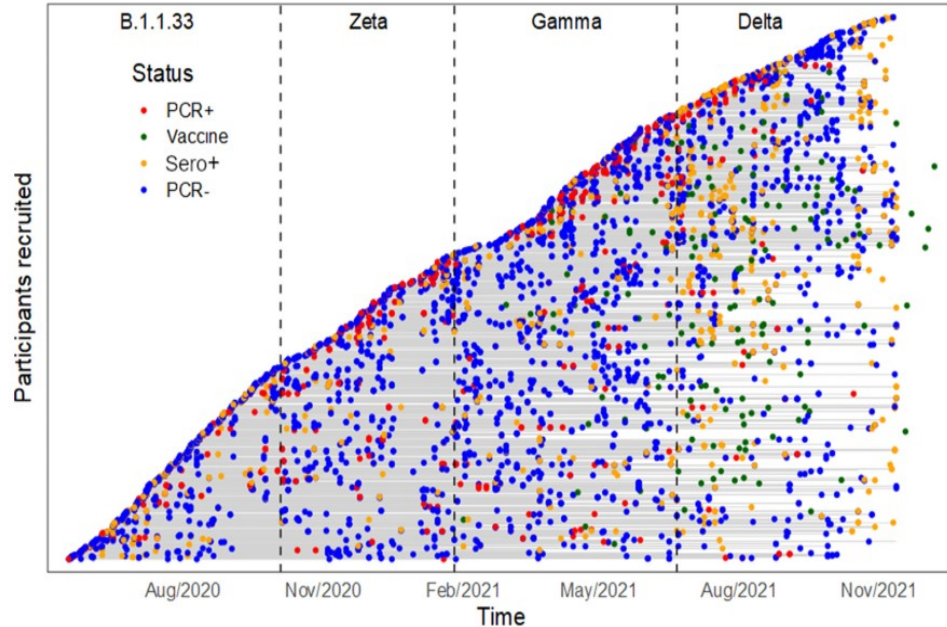


Figure 3 SARS-CoV-2 anti-S seroprevalence by month of sample collection.

Open access

Cohort profile

BMJ Open Cohort-profile: Household transmission of SARS-CoV-2 in a low-resource community in Rio de Janeiro, Brazil

Patrícia Brasil ¹, Luana Damasceno,¹ Trevon Fuller,^{1,2} Leonardo S Bastos,³ Oswaldo G Cruz,³ Fernando Medeiros,¹ Guilherme Amaral Calvet ⁴, Paola Resende,⁵ Jimmy Whitworth,⁶ Chris Smith ^{7,8}, Marilda M Siqueira,⁹ Marília Carvalho³

Uncertainty on seroprevalence studies (misclassification)

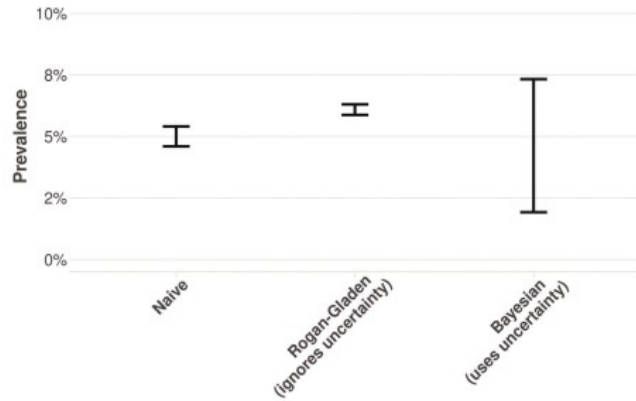


Figure 1 - Confidence (Naive and Rogan-Gladen) and credible (Bayesian) intervals for prevalence of COVID-19.



COMMENTS

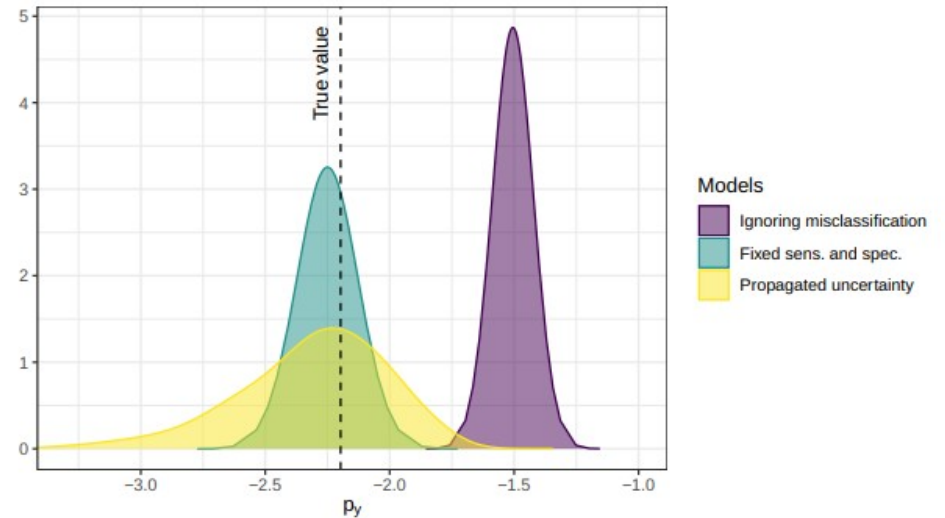
Sensitivity and specificity in prevalence studies: The importance of considering uncertainty

Rafael Izbicki¹, Márcio A. Diniz¹, Leonardo S. Bastos¹

¹Departamento de Estatísticas, Universidade Federal de São Carlos, São Carlos, SP, BR. ¹Programa de Computação Científica (PROCC), Fundação Oswaldo Cruz, Rio de Janeiro, RJ, BR.

Izbicki R, Diniz MA, Bastos LS. Sensitivity and specificity in prevalence studies: The importance of considering uncertainty. Clinics 2020;75:e2449

Izbicki, Diniz, Bastos(2020)



arXiv > stat > arXiv:2411.16311

Statistics > Methodology

[Submitted on 25 Nov 2024]

Bayesian models for missing and misclassified variables using integrated nested Laplace approximations

Emma Skarstein, Leonardo Soares Bastos, Håvard Rue, Stefanie Muff

Skarstein, Bastos, Rue, Muff (2024)

Incidence by COVID-19 wave (pre omicron)

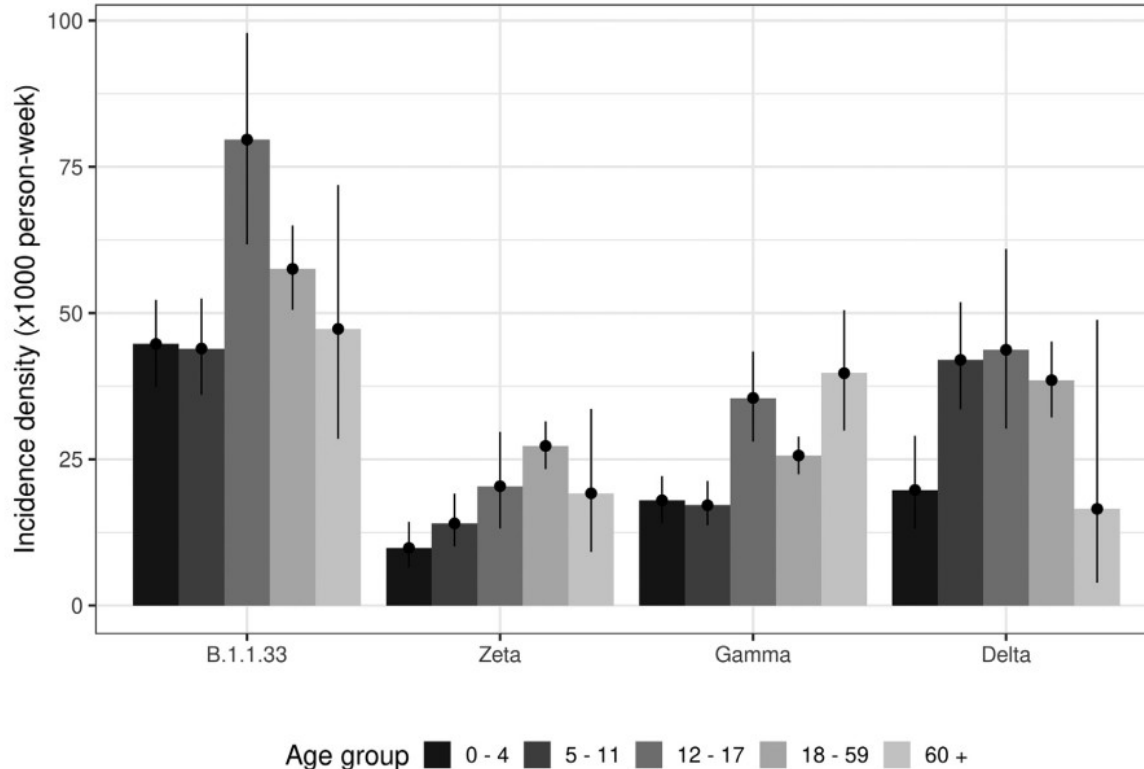


Figure 5. Incidence density by age group in each epidemic wave.

Incidence of SARS-CoV-2 over four epidemic waves in a low-resource community in Rio de Janeiro, Brazil: A prospective cohort study

Marília Sa Carvalho,^a Leonardo Soares Bastos,^a Trevon Fuller,^b Oswaldo Gonçalves Cruz,^c Luana Damasceno,^b Guilherme Calvet,^b Paola Cristina Resende,^c Chris Smith,^d Jimmy Whitworth,^d Marilda Siqueira,^a and Patricia Brasil^{b*}

^aScientific Computation Program, Oswaldo Cruz Foundation, Rio de Janeiro, RJ, Brazil

^bAcute Febrile Illnesses Laboratory, Evandro Chagas National Institute of Infectious Diseases, Oswaldo Cruz Foundation, Av. Brasil, 4365 - Manguinhos, Rio de Janeiro, RJ 21040-900, Brazil

^cLaboratory of Respiratory Viruses and Measles, Oswaldo Cruz Institute, Oswaldo Cruz Foundation, Brazil

^dDepartment of Clinical Research, London School of Hygiene and Tropical Medicine, London, United Kingdom

Summary

Background Incidence rates of SARS-CoV-2 infections in low-resource communities can inform vaccination strategies and non-pharmaceutical interventions (NPIs). Our objective was to estimate incidence over four epidemic waves in a slum in Rio de Janeiro, a proxy for economically deprived areas in the Global South.

Methods Prospective cohort of children and household contacts screened for SARS-CoV-2 by PCR and serology (IgG). The incidence density of PCR positive infections estimated for each wave - the first wave, Zeta, Gamma and Delta - was compared to an index combining NPIs and vaccination coverage.



The Lancet Regional Health - Americas
2022;12: 100283
Published online xxx
<https://doi.org/10.1016/j.lana.2022.100283>

Carvalho et al. (2022)

Incidence: refers to new cases

Schools: Should we have opened?

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Volume 148, Issue 1
July 2021



ARTICLES | JULY 01 2021

SARS-CoV-2 Infection Dynamics in Children and Household Contacts in a Slum in Rio de Janeiro **FREE**

Pâmella Lugon, MD; Trevon Fuller, PhD; Luana Damasceno, MS; Guilherme Calvet, MD, PhD; Paola Cristina Resende, PhD; Aline Rocha Matos, PhD; Tulio Machado Fumian, PhD; Fábio Correia Malta, MS; Aline Dessimoni Salgado, MD; Fernanda Christina Morone Fernandes, BS; Liege Maria Abreu de Carvalho, MD; Lusiele Guaraldo, PharmD, DSc; Leonardo Bastos, PhD; Oswaldo Gonçalves Cruz, PhD; James Whitworth, MD; Chris Smith, MD, PhD; Karin Nielsen-Saines, MD, MPH; Marilda Siqueira, PhD; Marília Sa Carvalho, MD, PhD; Patricia Brasil, MD, PhD

Lugon et al. (2021)



The Lancet Regional Health - Americas

Volume 5, January 2022, 100149



Comment

School reopening and COVID-19 in Brazil

Lorena G. Barberia ^a, Leonardo S Bastos ^b, Tatiane C Moraes de Sousa ^b

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<https://doi.org/10.1016/j.lana.2021.100149>

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Barberia, Bastos e Souza (2022)

What is the role of children in household transmission?

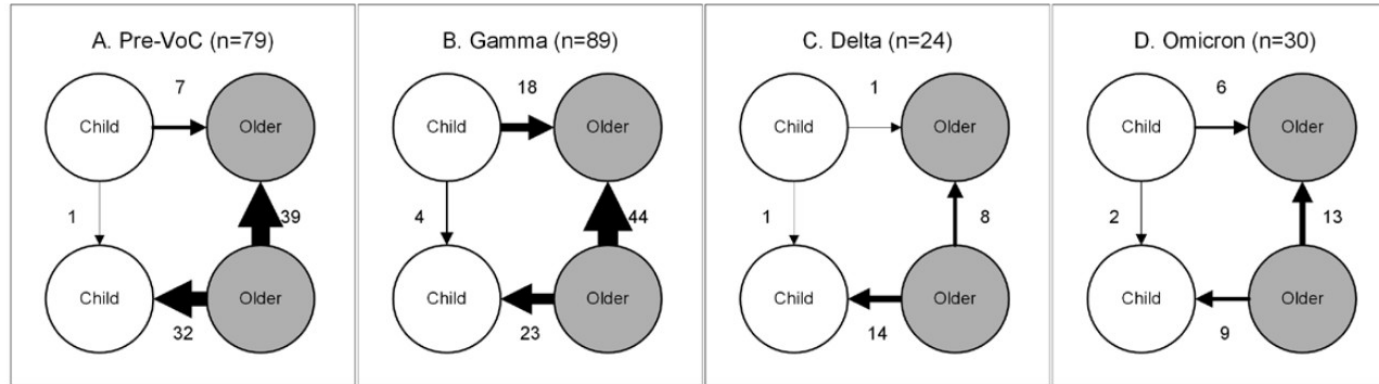


Figure 1. Number of household transmission events by the age of the index case and the secondary cases, stratified by the predominant SARS-CoV-2 lineage during the event.

Fuller, Bastos, et al. (2023)

Journal of the Pediatric Infectious Diseases Society

ORIGINAL ARTICLE



The Role of Children in Household Transmission of SARS-CoV-2 Across Four Waves of the Pandemic

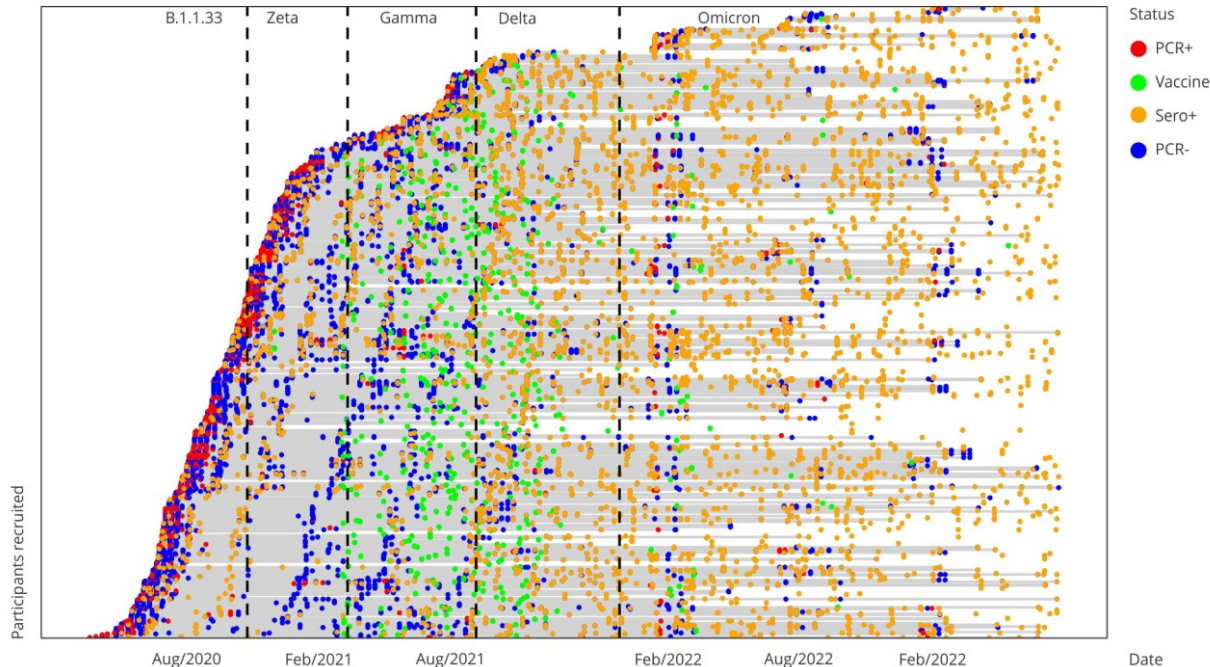
Trevon L. Fuller,^{1,*} Leonardo Bastos^{2,*}, Marília Sá Carvalho,³ Paola Cristina Resende,⁴ Luana Damasceno,⁵ Oswaldo Gonçalves Cruz,⁶ Fernando Medeiros,⁷ Guilherme Calvet,⁸ Lusiele Guarnaldo,⁹ Karin Nielsen-Saines,¹⁰ James Whitworth,¹¹ Christopher Smith,¹² Marilda Siqueira,¹³ and Patricia Brasil¹⁴

¹Department of Acute Febrile Illnesses, Evandro Chagas National Institute of Infectious Diseases, Oswaldo Cruz Foundation, Rio de Janeiro, Brazil, ²Scientific Computing Program, Oswaldo Cruz Foundation, Rio de Janeiro, Brazil, ³Laboratory of Respiratory Viruses and Measles, Oswaldo Cruz Institute, Oswaldo Cruz Foundation, Rio de Janeiro, Brazil, ⁴Division of Pediatric Infectious Diseases, David Geffen School of Medicine at UCLA, Los Angeles, California, USA, ⁵International Public Health, London School of Hygiene and Tropical Medicine, London, UK, and ⁶Clinical Research Department, London School of Hygiene and Tropical Medicine, London, UK

Open COVID-19 household cohort, Rio de Janeiro

Figure 2

Virological and immunological results for each cohort participant.



Cohort profile: follow-up of a household cohort throughout five epidemic waves of SARS-CoV-2 in Rio de Janeiro, Brazil

da Silva et al. (2024)

Families: 189

Participants: 691

Household transmission in Rio de Janeiro

	Pre VoC	Gamma/Delta	Omicron
Intercept-only Binomial GLMM			
Transmission probability (in cluster)	0.179 (0.128–0.229)	0.136 (0.084–0.186)	0.171 (0.134–0.214)
Binomial GLMM (Odds ratios)			
Variables			
Household density ¹	0.663 (0.332–1.321)	0.420 (0.078–2.266)	0.656 (0.225–1.911)
% Children under 12	0.898 (0.588–1.371)	1.650 (0.683–3.988)	1.345 (0.722–2.507)
% Adults over 60	1.222 (0.799–1.869)	0.727 (0.310–1.703)	1.041 (0.556–1.949)
Healthcare worker	0.689 (0.439–1.081)	2.227 (0.952–5.210)	0.486 (0.256–0.921)
SARS-CoV-2 infection-induced immunity score (average) ²	-	0.088 (0.023–0.341)	-
% of household members with hybrid immunity ³	-	-	0.095 (0.010–0.924)

Article | [Open access](#) | Published: 17 May 2025

Natural, vaccine-induced immunity and the probability of experiencing SARS-CoV-2 infection in a household cohort in Rio de Janeiro

[Michele F. B. da Silva](#), [Lusiele Guaraldo](#), [Leonardo Soares Bastos](#), [Heloisa F. P. Santos](#), [Otávio M. Espíndola](#), [Paola Cristina Resende](#), [Stephanie L. S. Penetra](#), [Anielle Pina-Costa](#), [Isabella C. V. de Moraes](#), [Guilherme A. Calvet](#), [Trevon L. Fuller](#), [Marilda M. Siqueira](#), [Marília Sá Carvalho](#) & [Patrícia Brasil](#) 

[Scientific Reports](#) 15, Article number: 17211 (2025) | [Cite this article](#)

da Silva et al. (2024)

- Vaccine-induced immunity protects against transmission.



What about reinfections?

Table 3. Risk Factors for SARS-CoV-2 Reinfection

Variable	Pre-Omicron	Omicron
People per room	0.59 (0.13–2.60)	0.99 (0.49–1.97)
Chronic rhinitis	2.93 (1.01–8.64)	1.06 (0.55–2.08)
Health care worker	2.58 (0.95–7.13)	0.70 (0.37–1.33)
Vaccination status: unvaccinated or partially vaccinated ^a	17.81 (5.72–56.91)	0.49 (0.17–1.44)

The model incorporated participant age and family and individual random effects to account for dependence among observations. The data indicate the relative risk of reinfection associated with each variable (+/- 95% credible interval). Variables in bold were associated with a significant increase in the risk of reinfection.

^aFully vaccinated or booster were aggregated and used as baseline in both pre-Omicron and Omicron models.

The Journal of Infectious Diseases

MAJOR ARTICLE

IDSA
Infectious Diseases Society of America

hivma
HIV medicine association

OXFORD

SARS-CoV-2 Reinfection Cases in a Household-Based Prospective Cohort in Rio de Janeiro

Stephanie L. S. Penetra,^{1,2} Heloisa F. P. Santos,¹ Paola Cristina Resende,² Leonardo Soares Bastos,² Michele F. B. da Silva,¹ Anielle Pina-Costa,¹ Renata Serrano Lopes,² Leonardo Saboia-Vahia,² Any Caroline Alves de Oliveira,² Elisa Cavalcante Pereira,² Fernando Medeiros Filho,^{1,3} Mayumi D. Wakimoto,¹ Guilherme A. Calvet,¹ Trevon L. Fuller,^{1,4} Jimmy Whitworth,^{5,6} Christopher Smith,^{5,6} Karin Nielsen-Saines,⁴ Marília Sá Carvalho,² Otávio M. Espindola,¹ Lusiele Guarnido,¹ Marilda M. Siqueira,² and Patricia Brasil^{1,2}

¹Evandro Chagas National Institute of Infectious Diseases, Oswaldo Cruz Foundation, Rio de Janeiro, Rio de Janeiro, Brazil; ²Laboratory of Respiratory Viruses and Measles National Influenza Centre, Americas Regional Reference Lab for Measles and Rubella, Reference Laboratory for COVID-19 World Health Organization, Oswaldo Cruz Institute, Oswaldo Cruz Foundation, Rio de Janeiro, Rio de Janeiro, Brazil; ³Scientific Computing Program, Oswaldo Cruz Foundation, Rio de Janeiro, Rio de Janeiro, Brazil; ⁴University of California, Los Angeles, Los Angeles, California, USA; ⁵International Public Health, London School of Hygiene and Tropical Medicine, London, United Kingdom; and ⁶School of Tropical Medicine and Global Health, Nagasaki University, Nagasaki, Japan

Penetra et al. (2023)

Data up to Aug. 2022

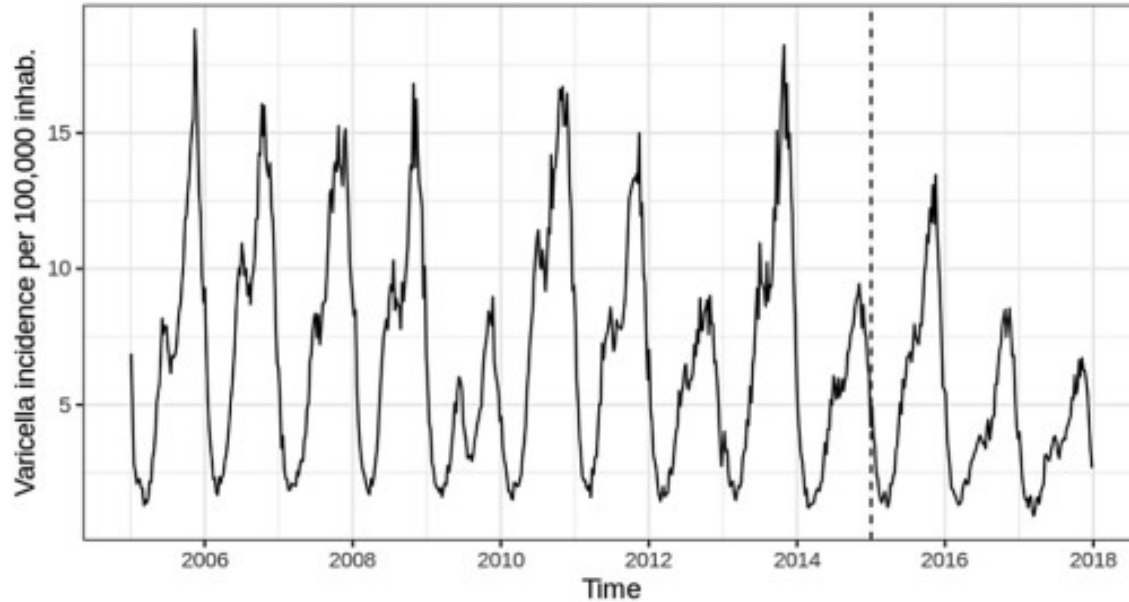
Vaccination offered protection in the Pre-omicron period!

No evidence during the first year of omicron.

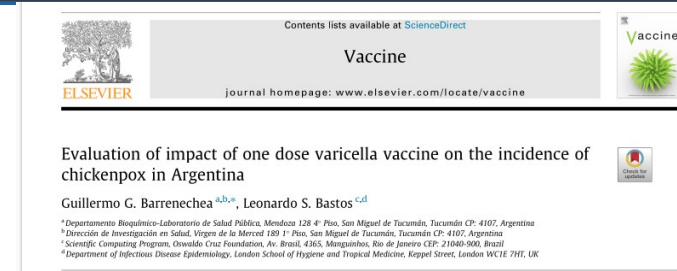
Modelling administrative data

- Evaluate interventions
- Early warning systems
- Learn about disease dynamics
- Attempt to forecast

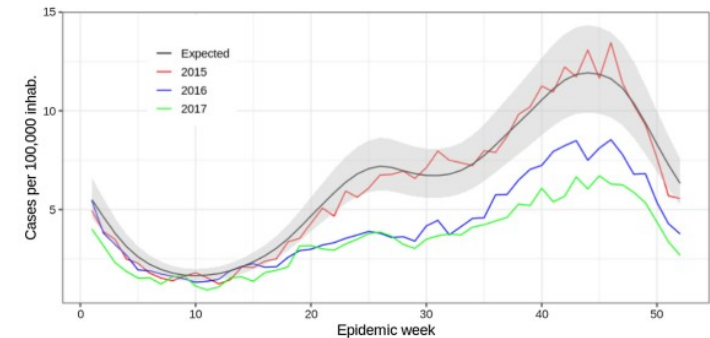
Impact of the one-dose varicella vaccine in Argentina



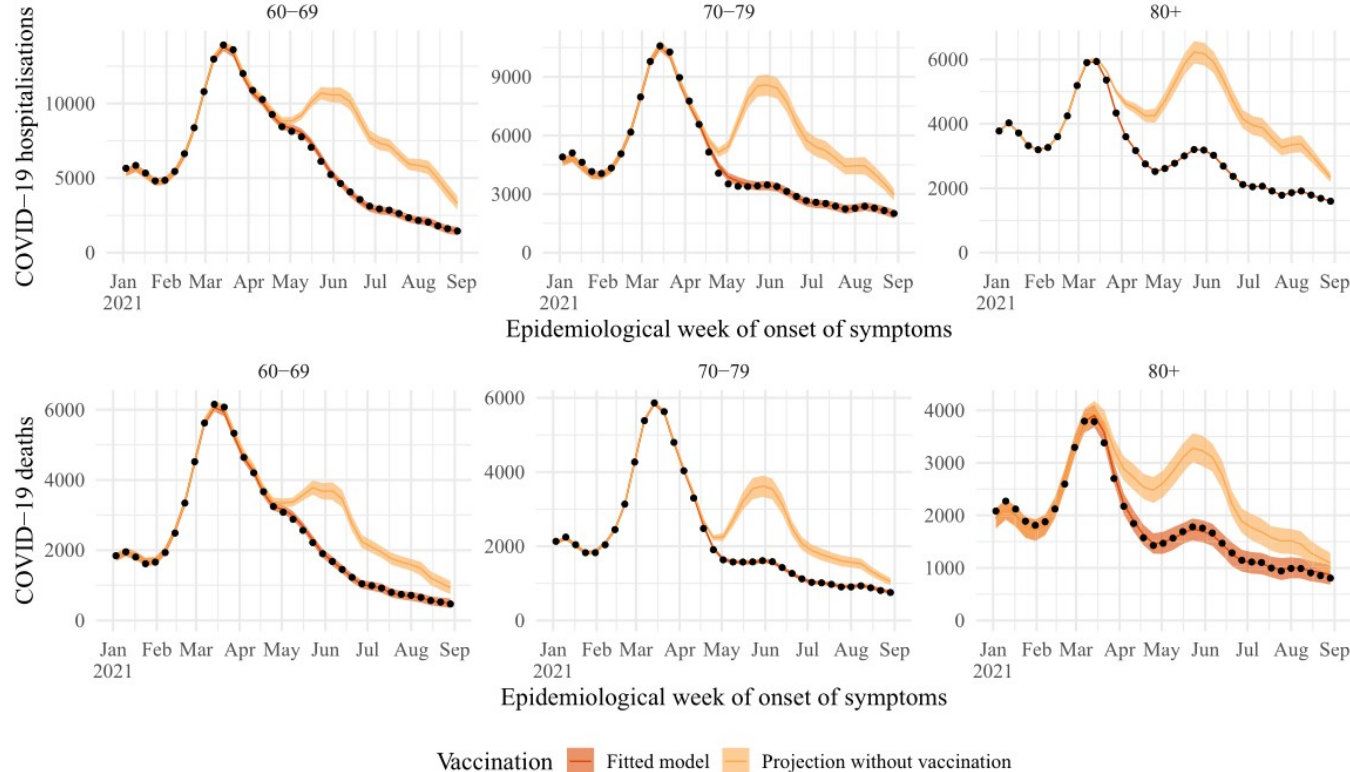
Negative Binomial mixed model with weekly structured and yearly iid random effects



Barrenechea & Bastos (2019)



Impact of COVID-19 vaccination: counterfactual analysis



The Lancet Regional Health - Americas
Volume 17, January 2023, 100397



Articles

Estimating the impact of implementation and timing of the COVID-19 vaccination programme in Brazil: a counterfactual analysis

Leonardo Souto Ferreira^{a, b}, Flavia Maria Darcie Marquitti^{b, c, d, e},
Rafael Lopes Paixão da Silva^{a, b}, Marcelo Eduardo Borges^b, Marcelo Ferreira da Costa Gomes^{b, d},
Oswaldo Gonçalves Cruz^{b, d}, Roberto André Kraenkel^{a, b}, Renato Mendes Coutinho^{b, e},
Paulo Inácio Prado^{b, f}, Leonardo Soares Bastos^{b, d}

Ferreira et al. (2023)



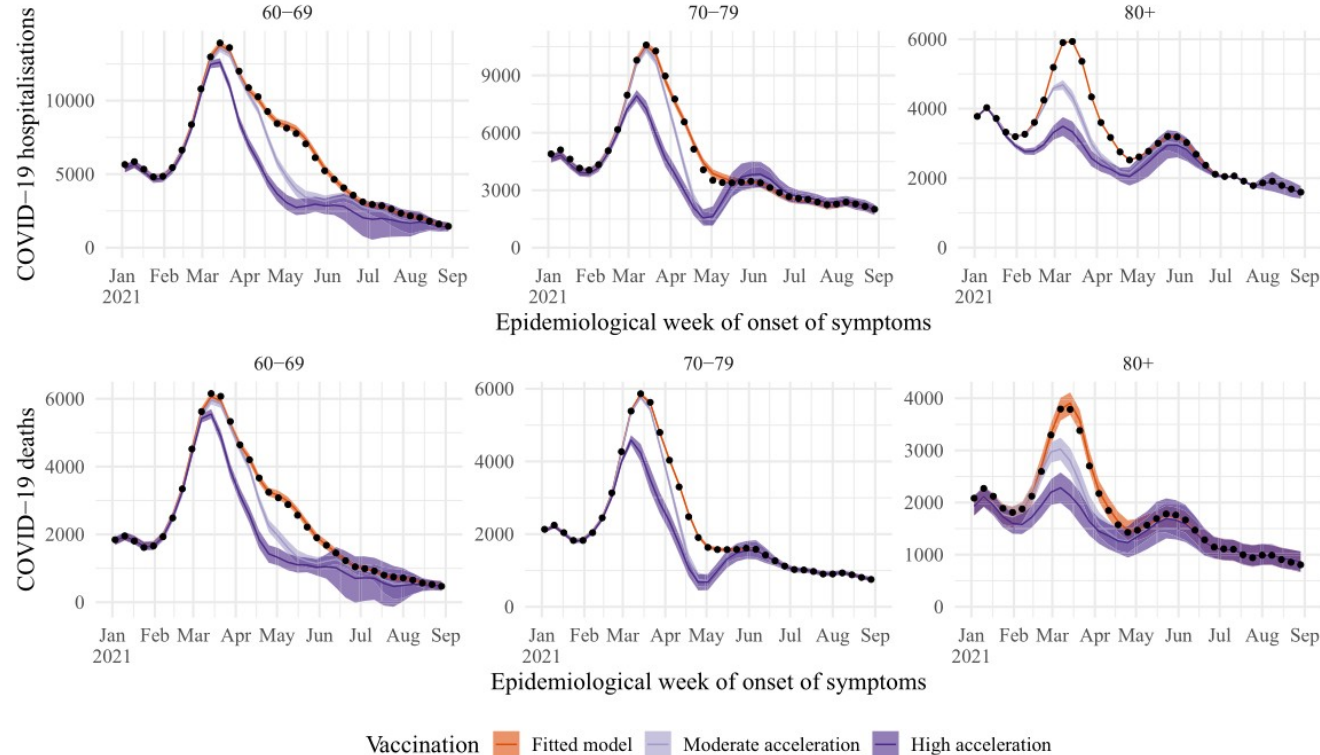
COMMENTS

How many hospitalizations has the COVID-19 vaccination already prevented in São Paulo?

Rafael Izbicki^{1, *}, Leonardo S. Bastos², Meyer Izbicki³, Hedibert F. Lopes⁴, Tiago Mendonça dos Santos^{5, 6}

Izbicki et al. (2021)

Impact of COVID-19 vaccination: counterfactual analysis



The Lancet Regional Health - Americas
Volume 13, September 2022, 100346



Comment

Government inaction on COVID-19 vaccines contributes to the persistence of childism in Brazil

Lorena G. Barberia ^a, Alexandra Boing ^b, Luciana Santana ^c, Wesley Cota ^d,
Leonardo S. Bastos ^e

Barberia et al. (2022)

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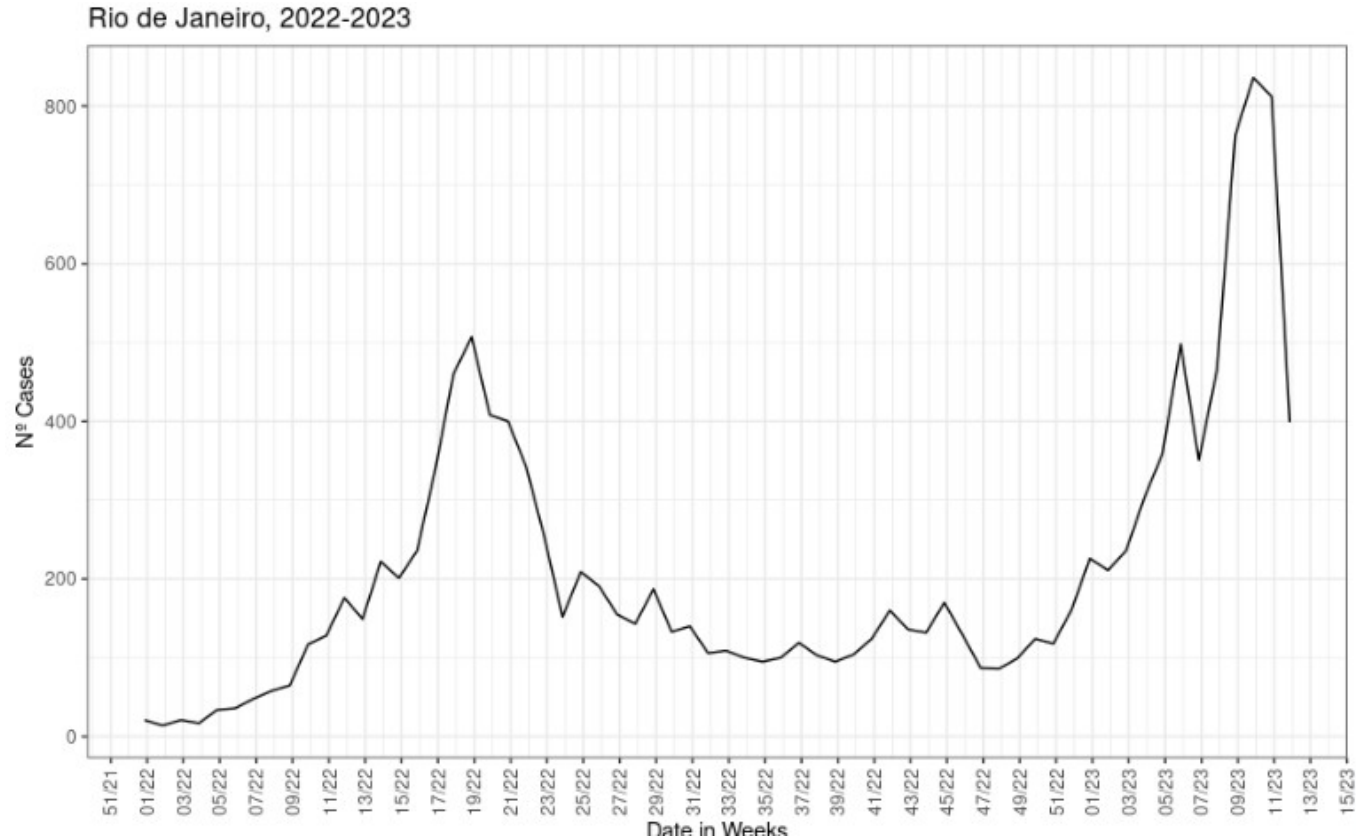
47 mil mortes de idosos por Covid seriam evitadas se vacinação começasse acelerada

Mais de 105 mil hospitalizações também seriam evitadas; mesmo assim, vacina reduziu mortes e complicações

Using administrative data for early warning

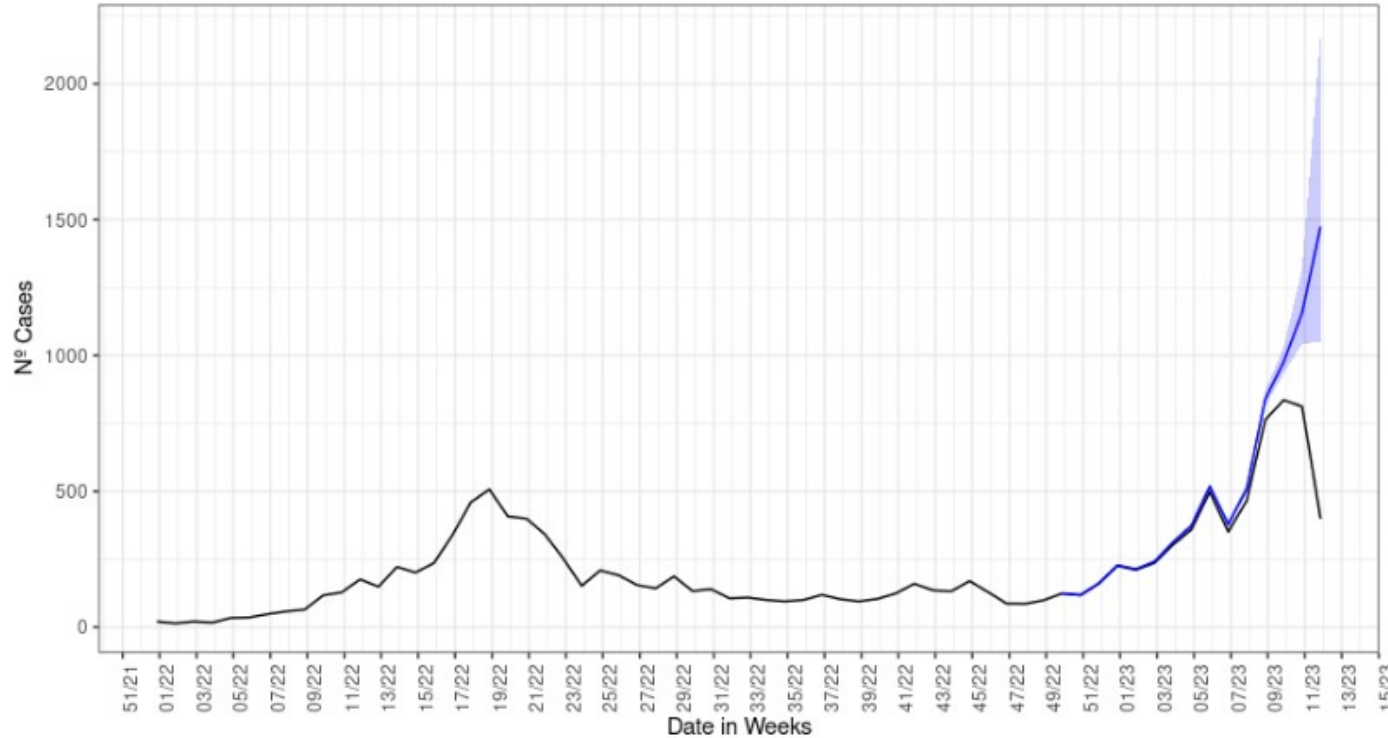
- **Notification of some diseases is compulsory**
 - e.g. dengue, chikungunya, SARI, malaria,...
 - **Note: Non hospitalized COVID-19 was never compulsory**
- **The most up-to-date inform suffers from delays**
 - Time between symptoms onset and the central system knows about the case.
 - Time between symptoms onset and search for assistance
 - Time between the case is identified and being “typed” into the system
 - Time to update the record (e.g. laboratory results)

Dengue cases in Rio, 2022-2023



Dengue cases in Rio, 2022-2023

Rio de Janeiro, 2022-2023



Statistics
in Medicine

RESEARCH ARTICLE | [Open Access](#) | [CC BY](#)

A modelling approach for correcting reporting delays in disease surveillance data

Leonardo S Bastos [✉](#), Theodoros Economou, Marcelo F C Gomes, Daniel A M Villela, Flavio C Coelho, Oswaldo G Cruz, Oliver Stoner, Trevor Bailey, Claudia T Codeço

First published: 10 July 2019 | <https://doi.org/10.1002/sim.8303> | Citations: 66

Bastos et al. (2019)

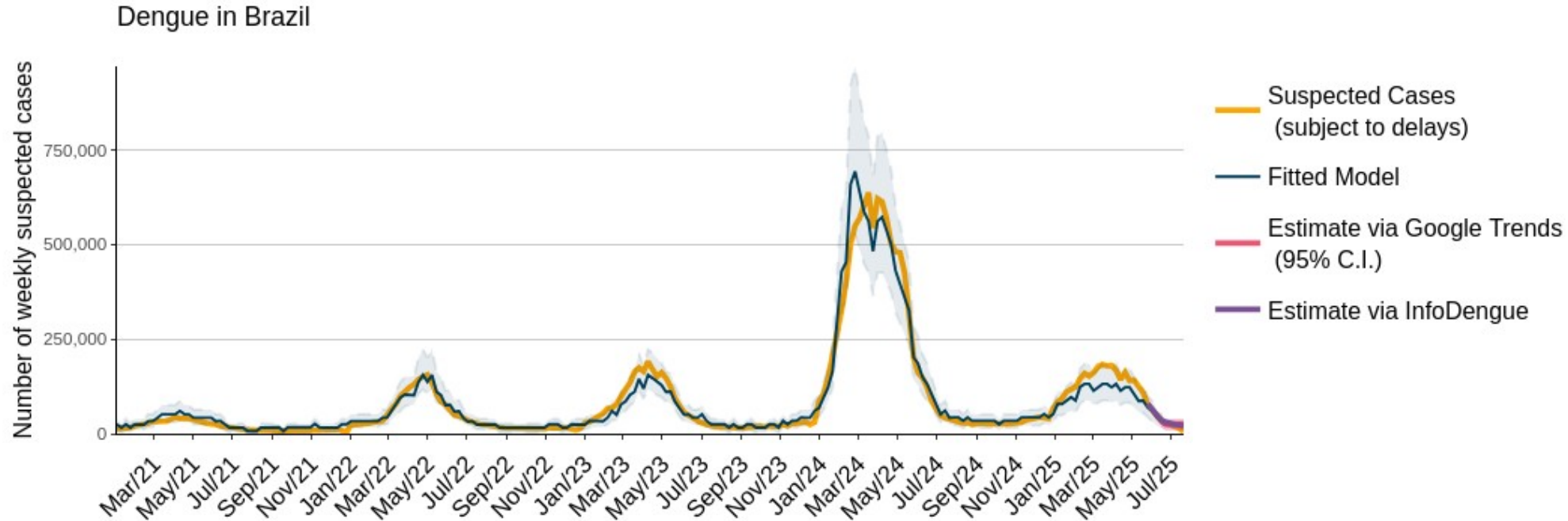
Chain-ladder model

Delay correction

Nowcasting

Nowcasting using google only

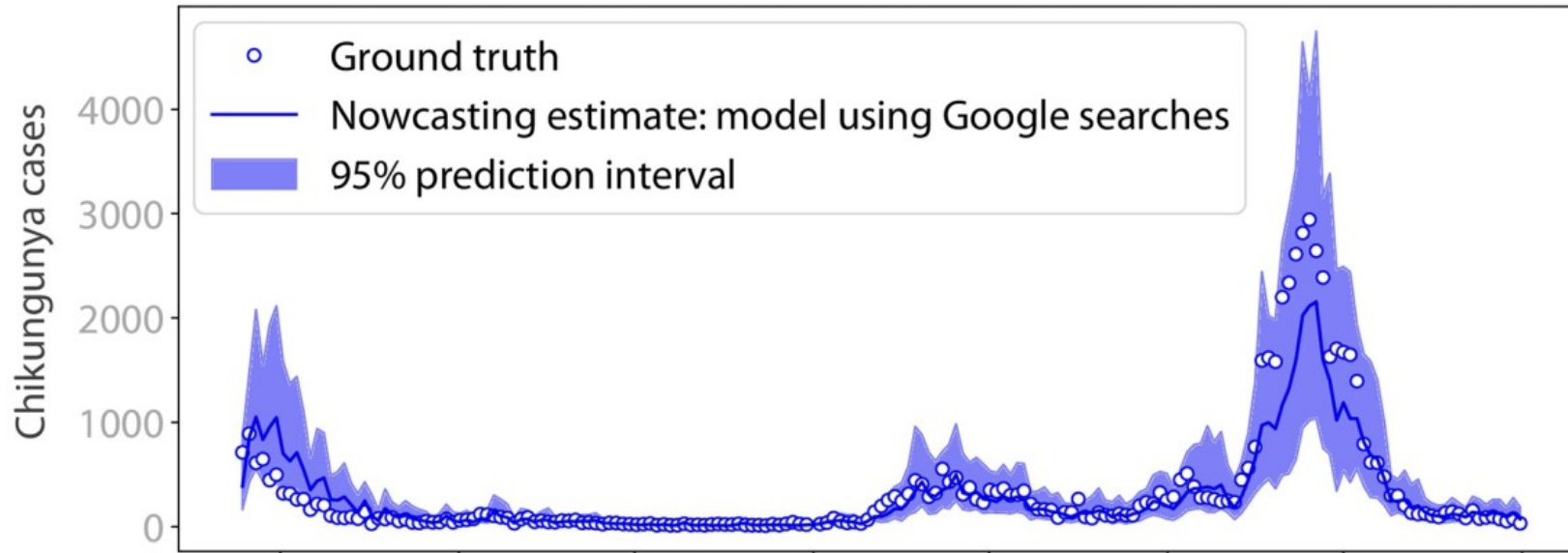
Corrected number of suspected dengue cases via Google Trends



Xiao et al. (2025, PLoS Negl Trop Dis)

Nowcasting using chain-ladder + google

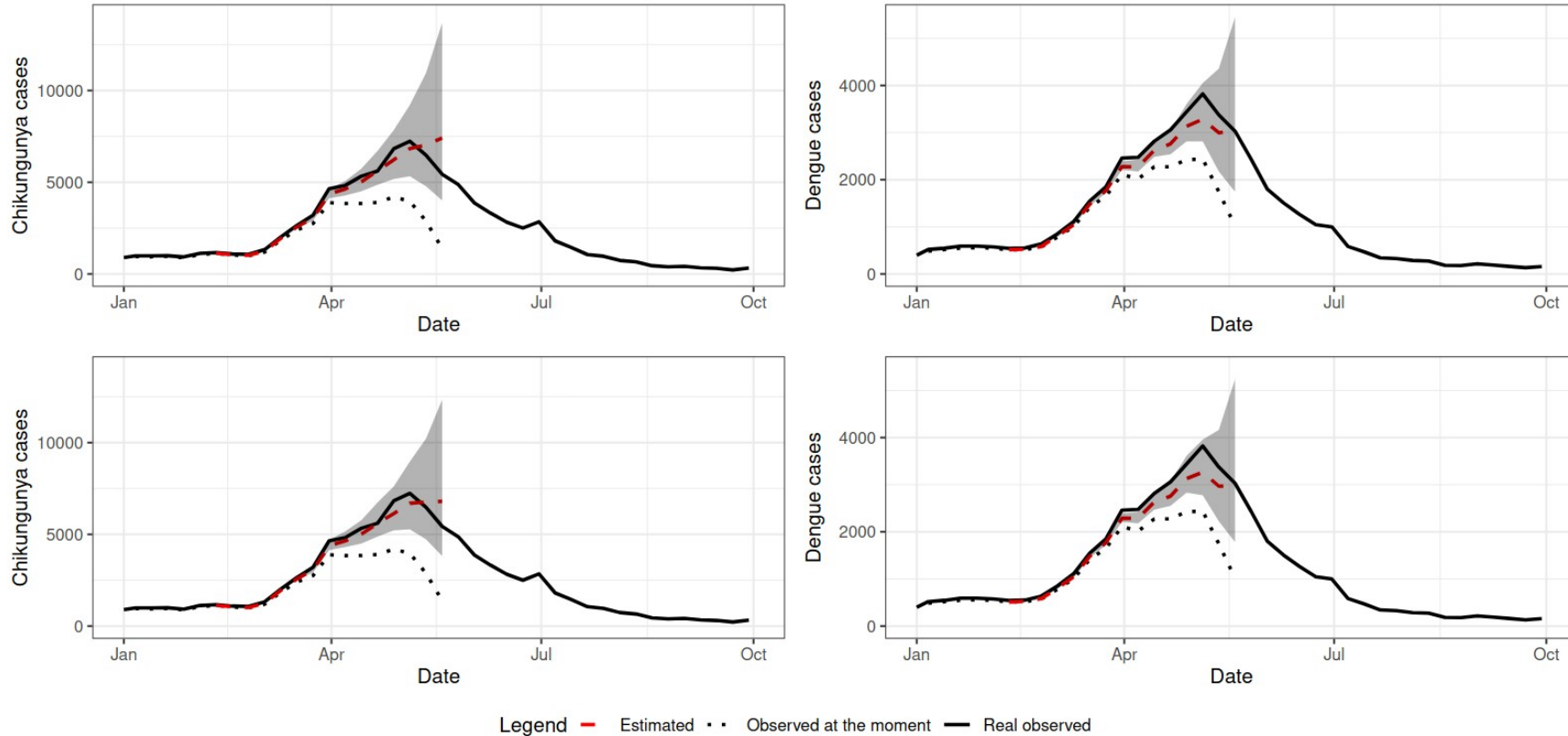
A



Bastos et al. (2019)

Miller et al. (2022, PLoS Negl Trop Dis)

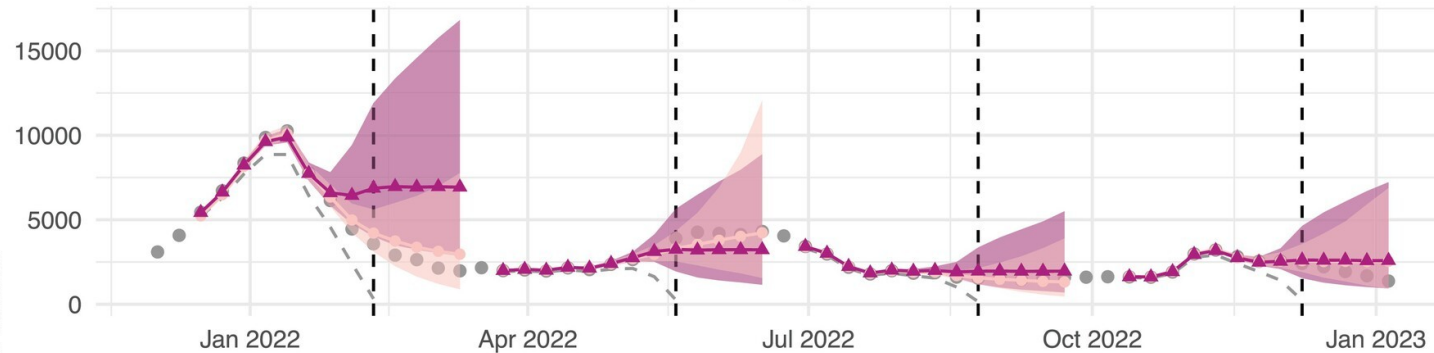
Multivariate nowcasting (dengue and chikungunya)



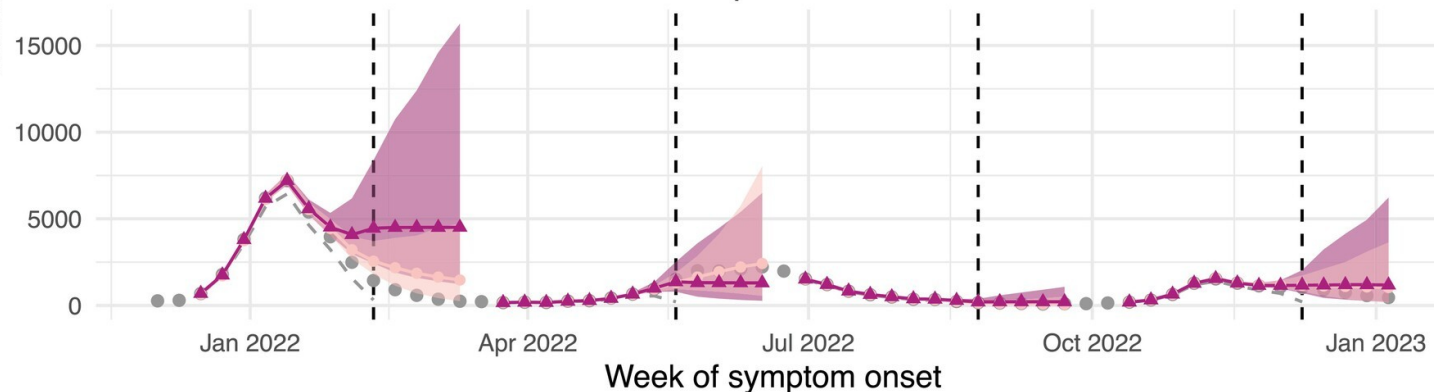
Santos (2023, M.Sc. UFRJ)

Joint model (SARI and SARI-COVID)

Severe acute respiratory illness in São Paulo



COVID-positive SARI



Haliday et al. (2026)



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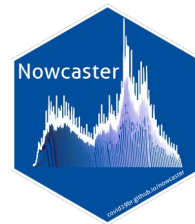
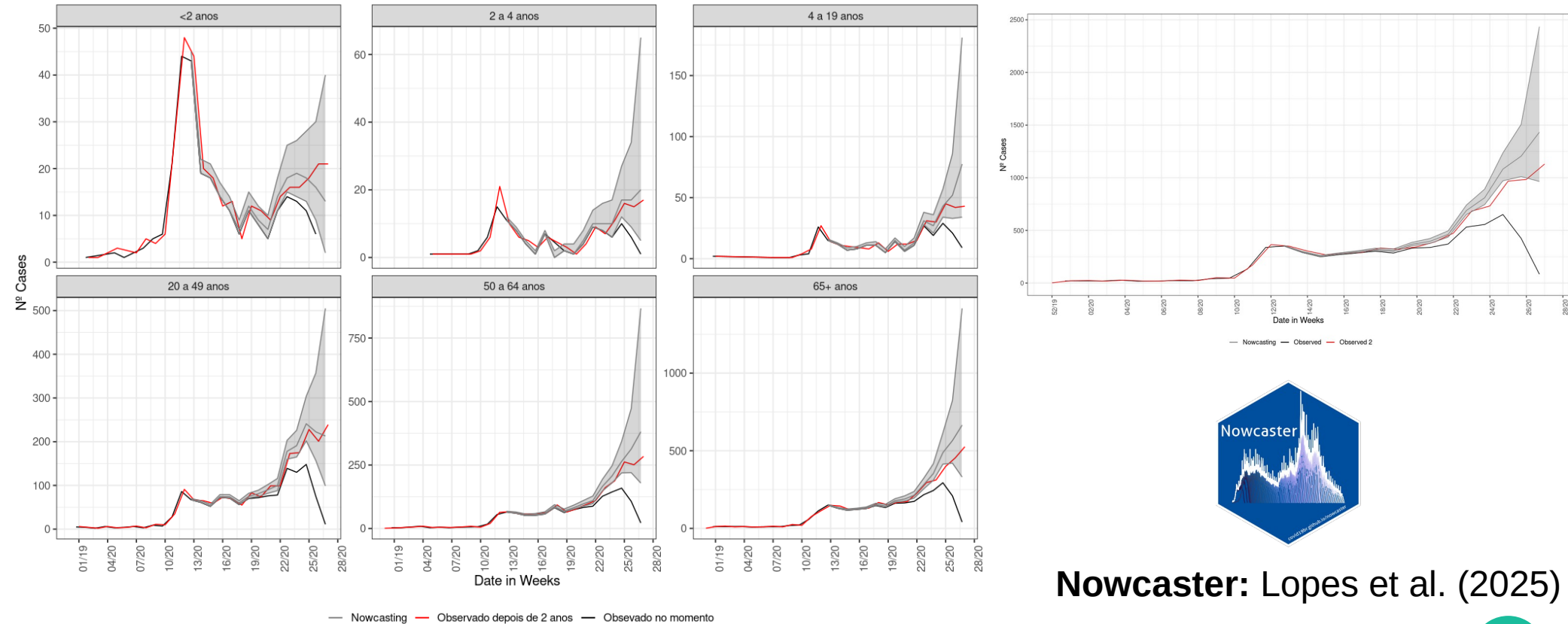
Joint Bayesian Nowcasting of Severe Acute Respiratory Illness and COVID-19 Positives in Brazil

[Alba Haliday](#) [Oliver Stoner](#) [Theo Economou](#) [Leonardo Soares Bastos](#)

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SARI cases by age (July 2020, Belo Horizonte)



Nowcaster: Lopes et al. (2025)

Vigilância da dengue: InfoDengue



<http://info.dengue.mat.br>

FGV EMap
ESCOLA DE
MATEMÁTICA
APLICADA

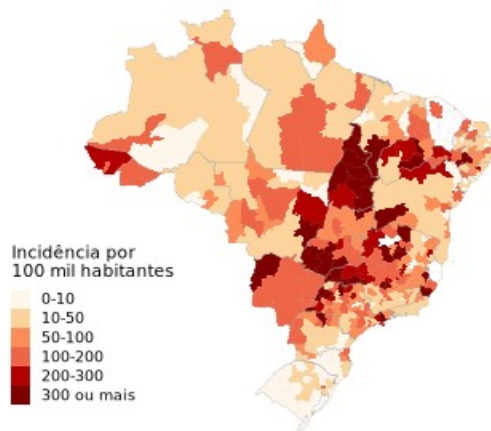


Home Data Report

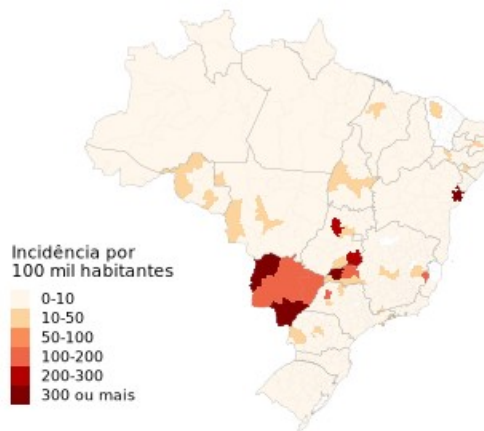
Find a municipality

Team About us

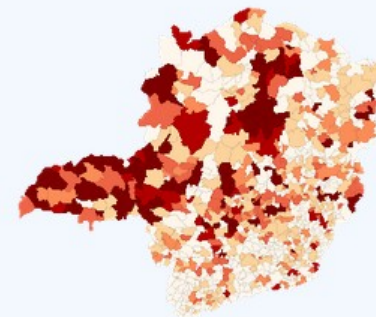
Dengue
SE 17-20/2026



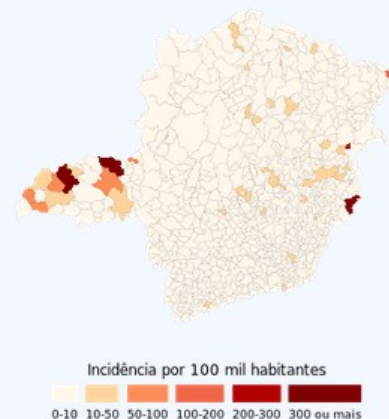
Chikungunya
SE 17-20/2026



Dengue
SE 17-20/2026



Chikungunya
SE 17-20/2026



Dengue e chikungunya em BH (InfoDengue)



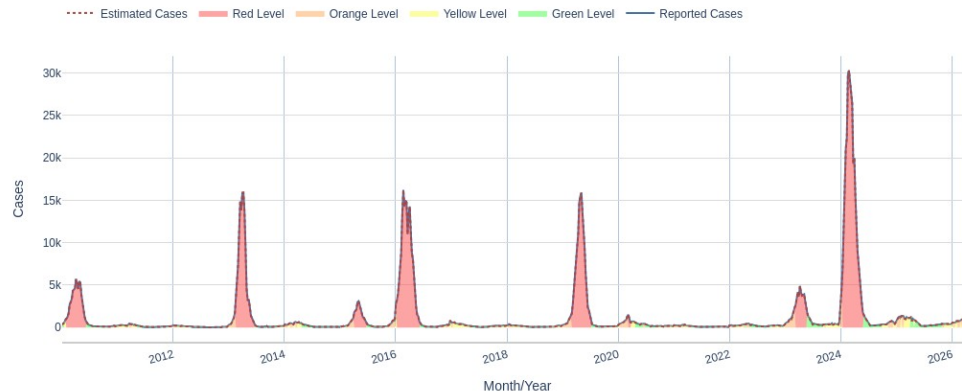
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Dengue Situation - Belo Horizonte at May 23, 2026

- **Population:** 2.4 million of inhabitants.
- **Estimated Incidence epidemiological week 20:** 22.7 per 100 thousand inhabitants.
- **Arborivirosis Situation report of :** [Belo Horizonte](#)
- **For more details go to the States page:** [MG](#)

Select the disease to be shown ☒ Dengue ☐ Chikungunya ☐ Zika

Chance of alert orange or red for Dengue in the municipality of Belo Horizonte in the next week: 3.4%



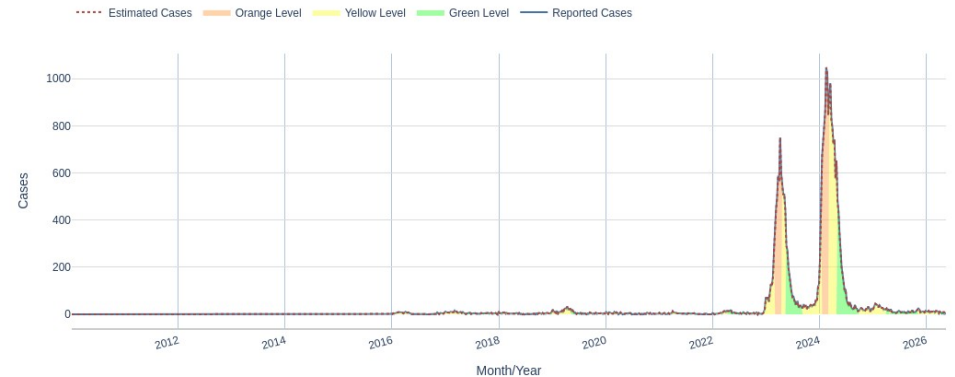
Home Data Report Find a municipality Team About us

Chikungunya Situation - Belo Horizonte at May 23, 2026

- **Population:** 2.4 million of inhabitants.
- **Estimated Incidence epidemiological week 20:** 0.4 per 100 thousand inhabitants.
- **Arborivirosis Situation report of :** [Belo Horizonte](#)
- **For more details go to the States page:** [MG](#)

Select the disease to be shown ☐ Dengue ☒ Chikungunya ☐ Zika

Chance of alert orange or red for Chikungunya in the municipality of Belo Horizonte in the next week: 62.2%

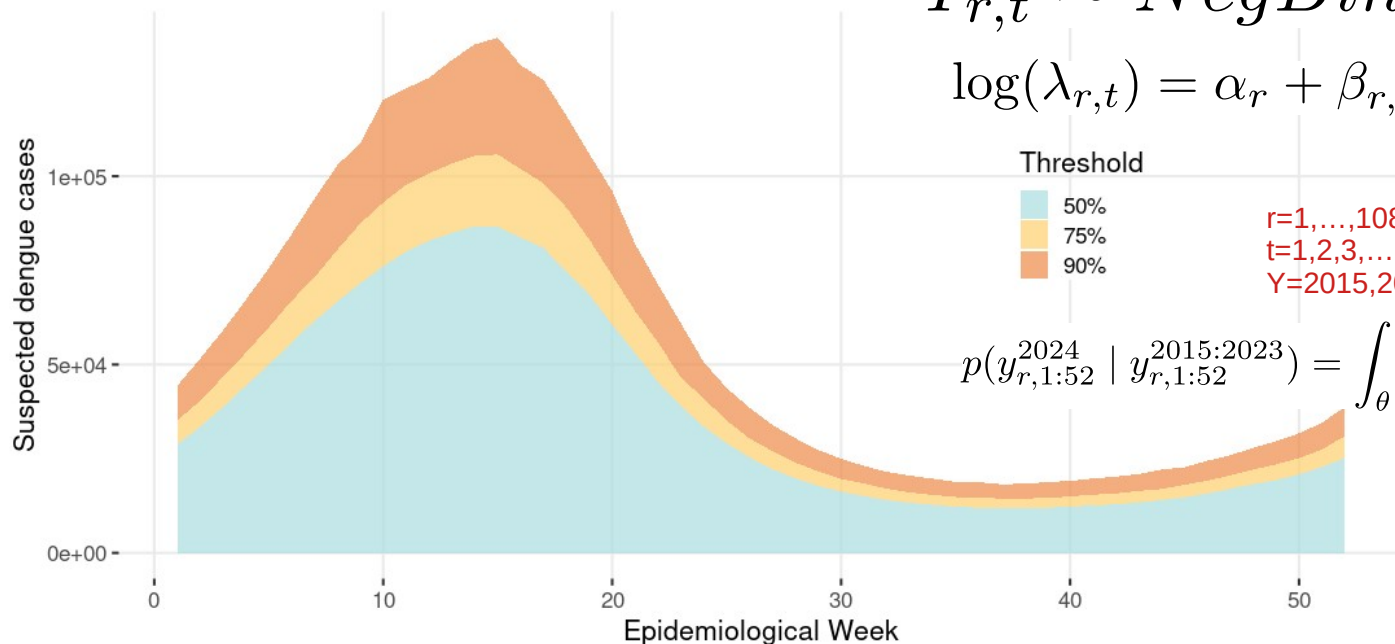


É possível prever a dengue? (Dezembro de 2023)

- Em 2023 (ano de El Nino), o ministério da saúde nos pediu uma previsão de dengue para 2024

$$Y_{r,t} \sim NegBin(\lambda_{r,t}, \phi_r)$$

$$\log(\lambda_{r,t}) = \alpha_r + \beta_{r,W[t]} + \gamma_{r,Y[t]}$$



$$p(y_{r,1:52}^{2024} | y_{r,1:52}^{2015:2023}) = \int_{\theta} p(y_{r,1:52}^{2024} | \theta) p(\theta | y_{r,1:52}^{2015:2023}) d\theta$$

Total de casos (prev):
1.46M [0,96 ; 4.22]

Janeiro de 2024

g1

SAÚDE

Brasil pode chegar a 4,2 milhões de casos de dengue em 2024, diz Ministério da Saúde

Os primeiros 521 municípios a receberem a vacina contra a doença devem ser cobertos até o final de março. Crianças de 10 a 11 anos serão primeiro grupo a receber o imunizante.

Por **Paloma Rodrigues**, TV Globo
09/02/2024 08h34 · Atualizado há um ano



agênciaBrasil

ÚLTIMAS NOTÍCIAS | CULTURA | DIREITOS HUMANOS | ECONOMIA | EDUCAÇÃO | ESPORTES | GERAL | INTERNA

PREPARAÇÃO

- Preparação mais assertiva
- Reunião preparatória para a alta transmissão de dengue
- Planejamento compra de insumos e abertura de processos de compra de vacinas, reidratação oral, soro, paracetamol, dipirona, kits inseticidas...
- Notas de Alerta aos estados e municípios no final de 2023
- Implementação da Sala Nacional de Arbovírus, em Dezembro de 2023

RESPOSTA

- Resposta mais direcionada
- Compra de vacinas
- Transferência de recursos
- Capacitação de multiplicadores no manejo clínico da dengue e chikungunya
- Abertura do Centro de Operações de Emergências (COE) para dengue e outras arboviroses

Saúde

Brasil pode registrar até 4,2 milhões de casos de dengue em 2024

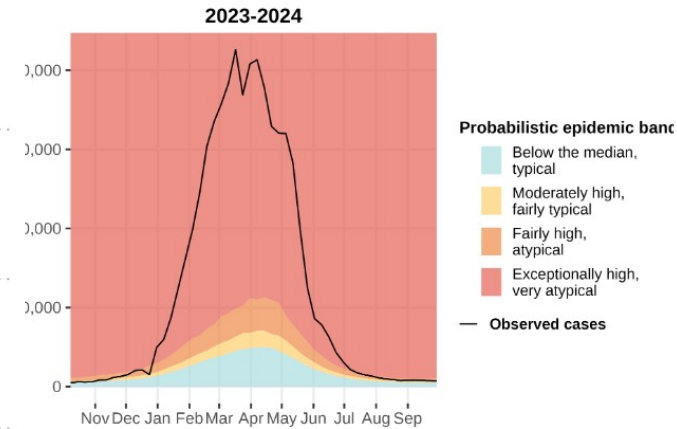
Estimativa é do Ministério da Saúde

PAULA LABOISSIÈRE – REPÓRTER DA AGÊNCIA BRASIL

Publicado em 30/01/2024 - 12:24
Brasília



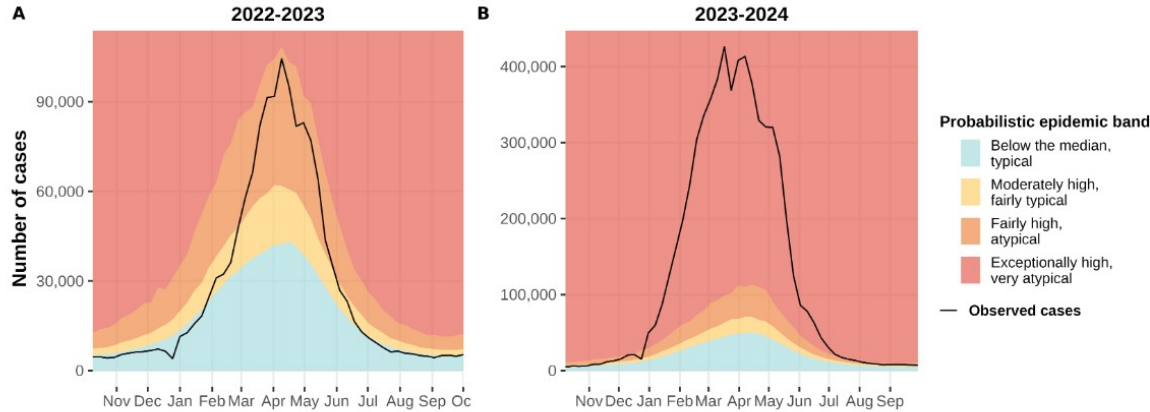
O que aconteceu?



Predito:
1.46M [0,96 ; 4.22]

Observado:
6.4M (!)

Bandas epidêmicas probabilísticas



Faixa	Nível dos casos	Interpretação
$\leq 50\%$	Abaixo da mediana, típico	Número de casos abaixo da mediana histórica, com 50% de probabilidade de ter ocorrido no passado.
50-75%	Moderadamente alto, bastante típico	Número de casos moderadamente maior que a mediana histórica, com 25% de probabilidade de ter ocorrido no passado.
75-90%	Bastante alto, atípico	Número de casos bastante acima da mediana histórica, com 15% de probabilidade de ter ocorrido no passado.
$\geq 90\%$	Excepcionalmente alto, muito atípico	Número de casos excepcionalmente acima da mediana histórica, com 10% de probabilidade de ter ocorrido no passado.



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Buscar na Fiocruz

Você está aqui: Início / Notícias / Pesquisadores Da Fiocruz Vencem Prêmio Com Modelo Que Prevê Epidemias de Dengue

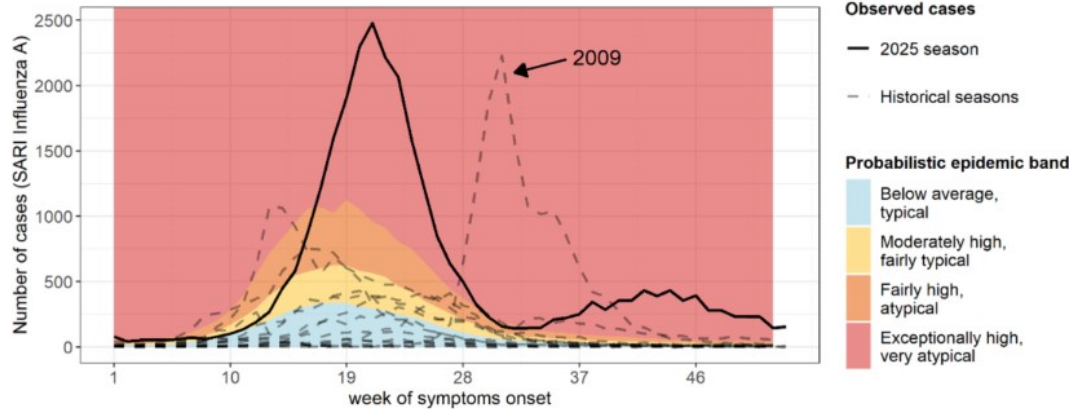
Pesquisadores da Fiocruz vencem prêmio com modelo que prevê epidemias de dengue

Por Luisa Picanço (VPEC/Fiocruz)
Publicado em 04/05/2026 14:14

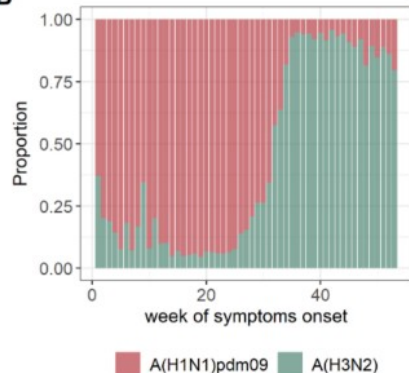


Descrevendo a epidemia de Influenza em 2025

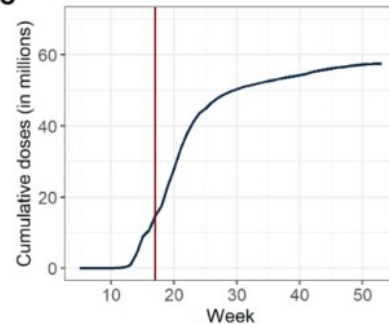
A



B



C



Portela et al. (2026)

Influenza and other respiratory viruses

ISRV
International Society for
Respiratory Viruses

Forward
Series

LETTER TO THE EDITOR | [Open Access](#) | [CC](#) [BY](#) [NC](#) [ND](#)

Highly Atypical 2025 Influenza A Season in Brazil

[Tatiana P. Portella](#) [✉](#) [Laís Picinini Freitas](#) [Paola Cristina Resende](#) [Marcelo Ferreira da Costa Gomes](#)
[Leonardo S. Bastos](#) [✉](#)

First published: 22 April 2026 | <https://doi.org/10.1111/irv.70262> | [VIEW METRICS](#)



Vigilância de SRAG: InfoGripe

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A FUNDAÇÃO PESQUISA E ENSINO SERVIÇOS PRODUÇÃO E INOVAÇÃO

Você está aqui: Início / Notícias / InfoGripe Aponta Aumento Do Número de Casos de SRAG Em Crianças Em Alguns Estados

InfoGripe aponta aumento do número de casos de SRAG em crianças em alguns estados

Por Regina Castro e Rhyann de Meira (estágio supervisionado por Raquel Aguiar)
Publicado em 31/07/2025 15:16

Boletim InfoGripe: aponta aumento de casos de SRAG em crianças pequenas

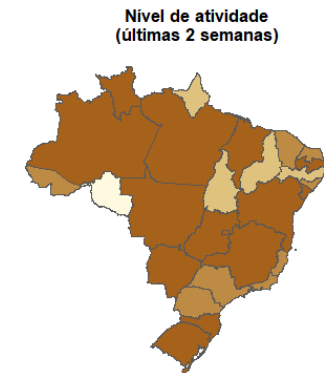
From a government health agency of Brazil

BOLETIM INFOGRIFE

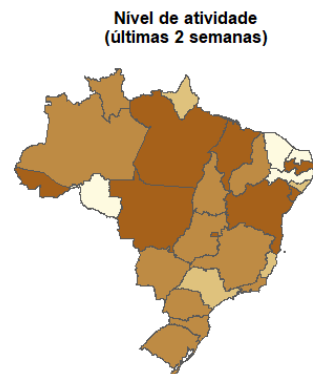
AUMENTO DE CASOS DE SRAG EM CRIANÇAS PEQUENAS

GRAVADO EM 31/07

Watch on YouTube



INFO GRIPE

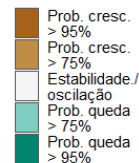


INFO GRIPE

**Semana 20 2026
(17/05 - 23/05):
Estados e DF**



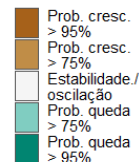
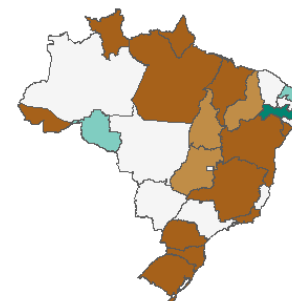
**Tendência de longo prazo
(últimas 6 semanas)**



**Capitais e região
central de saúde do DF**

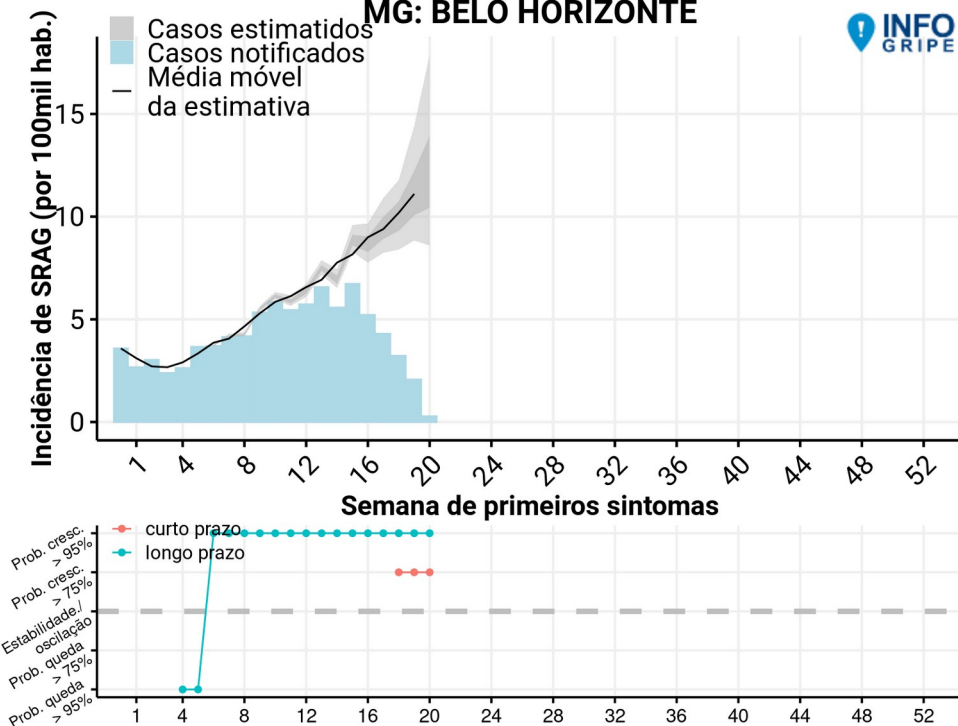


**Tendência de longo prazo
(últimas 6 semanas)**



InfoGripe: SRAG em BH (Até a 20ª epiweek)

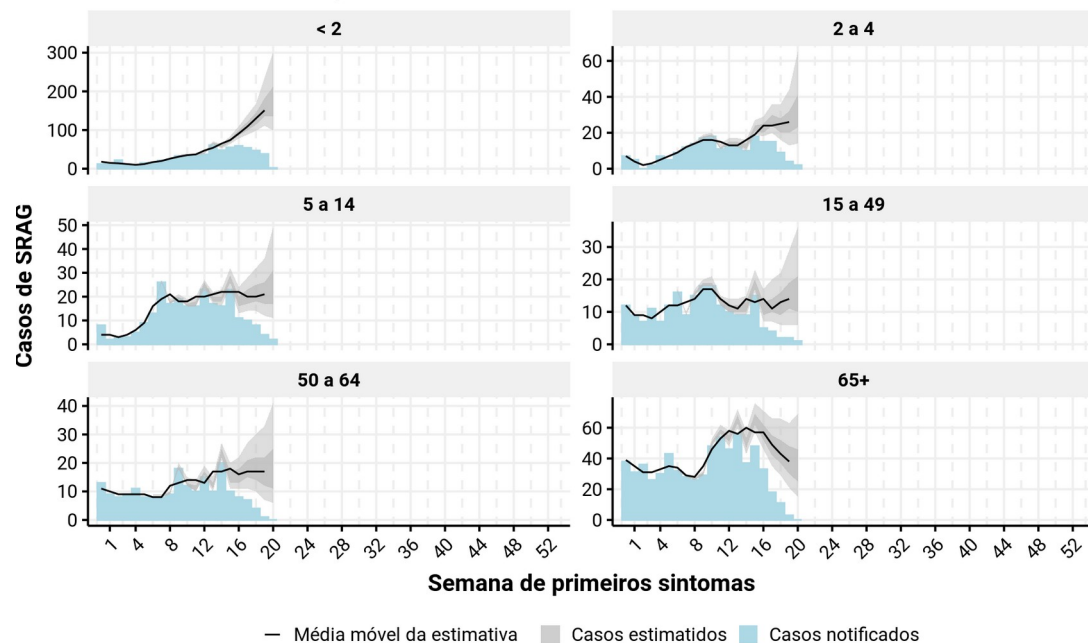
MG: BELO HORIZONTE



MG: BELO HORIZONTE



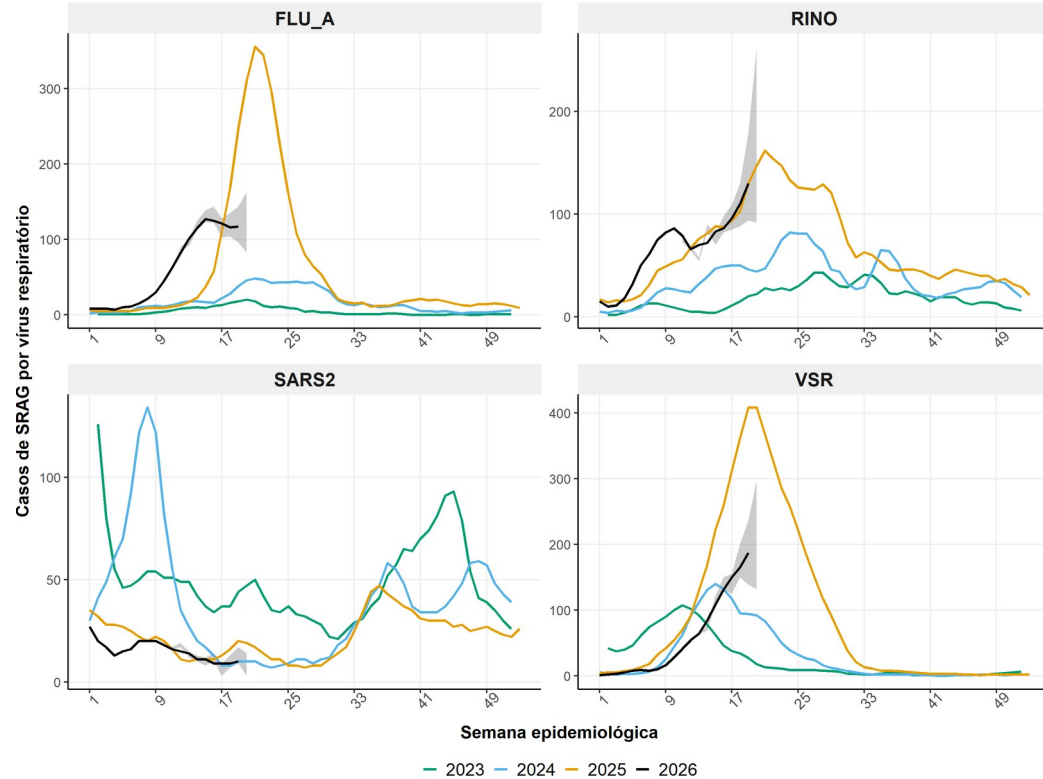
Novos casos semanais por faixa etária. Dados até a semana 20 2026



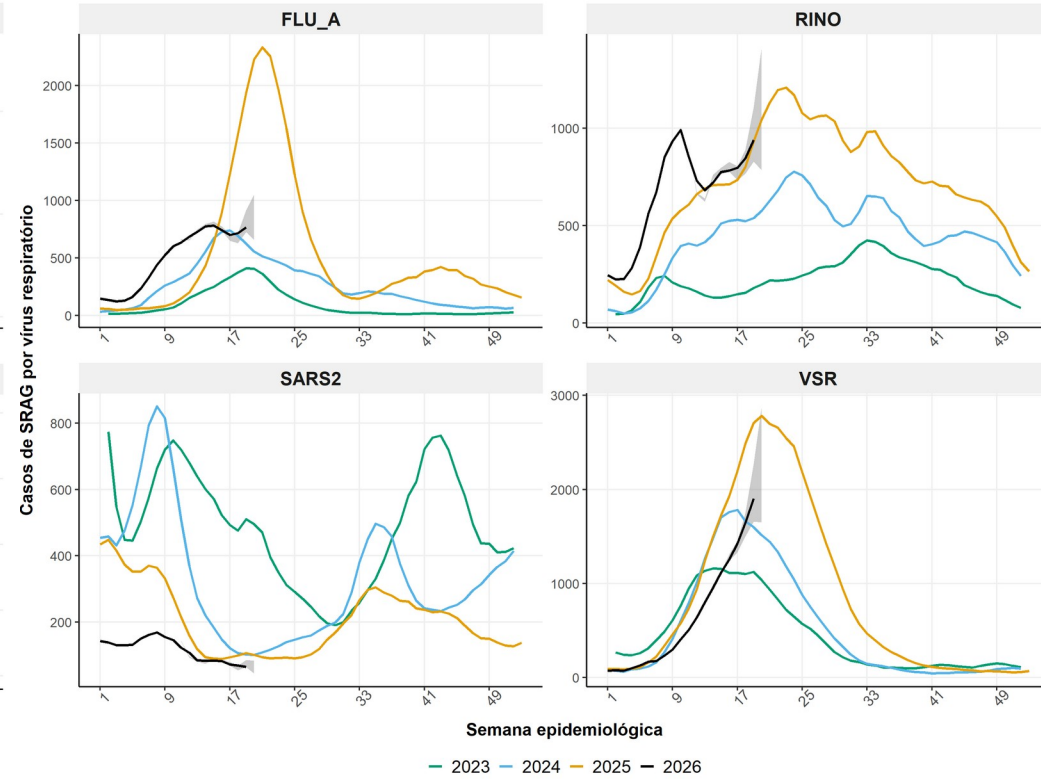
Provável causador do cenário atual **VSR (vírus sincicial respiratório)**

SRAG por vírus (Work in progress)

MG - Casos estimados e notificados de SRAG até a Semana 20 de 2026



BR - Casos estimados e notificados de SRAG até a Semana 20 de 2026



Conclusions



- **Bayesian methods are already been used by some surveillance teams in Brazil**
 - and incorporated in the decision-making process!
- **Computational cost is very important to be considered**
 - INLA offers an efficient solution for a class of models (**Exploring GAM!**)
- **There is still a large number of challenges in order to model infectious diseases data**
 - Population immunity; climate change/climate events; different interventions;



Obrigado!



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<https://lsbastos.github.io>

